Interpretable Movie Success Prediction Using Movie Personas and Ensemble Models

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Code Repository: https://github.com/nivedhasundar04/movie-success-prediction.git

Abstract

Predicting commercial success in the film industry is challenging due to the heterogeneous nature of movies across budget levels, genres, and audiences. Using 11,496 commercially released TMDB films, we apply K-Means clustering to identify three movie personas (Mainstream Blockbusters, Mid-Budget Commercial, and Indie/Low-Visibility films) and compare three experimental setups across Logistic Regression, Random Forest, and XGBoost. The global XGBoost model achieves the highest ROC-AUC of 0.837, with persona-aware models performing within 0.017 of this baseline. However, SHAP analysis reveals that success drivers differ meaningfully across personas: budget dominates for Mainstream Blockbusters, audience reach drives Indie films, and rating quality is most predictive for Mid-Budget Commercial productions. These findings suggest that persona-aware analysis provides actionable interpretability even when a global model is sufficient for prediction.

1 Introduction

This project investigates whether segmenting movies into interpretable personas improves success prediction and whether success drivers differ across film types. Using a pipeline combining K-Means clustering, XGBoost classification, and SHAP interpretability analysis, movies are grouped into Blockbuster, Indie/Niche, and Mid-Tier personas. Three experimental configurations are compared: a global baseline, a cluster-augmented global model, and fully separate per-persona models, evaluated across Logistic Regression, Random Forest, and XGBoost. The project emphasizes both predictive performance and interpretability, examining how budget, IMDb votes, ratings, popularity, release timing, and genre influence success differently across film types.

2 Related Work

Research on predicting movie commercial success spans economics, information systems, and machine learning. Prior work can be grouped into three conceptual categories: feature-based prediction models, clustering and segmentation approaches, and hybrid explainable frameworks that combine both.

2.1 Feature-Based Prediction Models

Early work by Litman (1) established that production budget, star power, genre, and distributor identity are significant predictors of box office revenue. Subsequent studies extended this foundation using regression and classification frameworks. De Vany and Walls (2) demonstrated that movie revenues follow power-law distributions, complicating standard regression assumptions and motivating the use

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of classification-based targets, such as the binary success label we define as revenue exceeding twice the production budget.

Tree-based ensemble methods have since become the dominant approach for tabular movie prediction tasks. Quader et al. (4) applied Random Forest and Gradient Boosting to TMDB metadata, reporting ROC-AUC scores in the range of 0.78 to 0.84, which is consistent with the baseline performance observed in our global experiments. XGBoost has been specifically favored in recent work for its strong performance on structured tabular data with mixed feature types (9), and we adopt it as our primary classifier for the same reasons. Logistic Regression was retained as a linear baseline, consistent with the approach taken by (10) and the example projects in this course, where simpler models provide a reference for evaluating the marginal contribution of more complex ones.

2.2 Clustering and Segmentation Approaches

Unsupervised learning methods such as K-Means and hierarchical clustering have long been applied for market segmentation. Li and Li (8) combined K-Means with PCA for customer behavior segmentation, finding that cluster-aware analysis surfaces behavioral patterns invisible to global models. Wedel and Kamakura (7) provide a foundational treatment of heterogeneous market segmentation, and more recent work has applied similar ideas to digital content and streaming platforms.

In the movie domain, Sharda and Delen (3) segmented films by budget tier before applying separate neural network classifiers, finding moderate performance improvements for high-budget and low-budget segments. However, their work did not formally test whether segmentation-aware models outperform global models under controlled experimental conditions. Our project addresses this gap directly. We apply K-Means clustering over the full Lasso-selected feature set, use both the Elbow Method and Silhouette Score to select k=3, and identify three interpretable personas: Mid-Budget Commercial, Mainstream Blockbusters, and Indie/Low-Visibility films.

2.3 Hybrid and Explainable Frameworks

To connect segmentation and prediction, several hybrid pipelines have been proposed. Ghiassi et al. (5) developed a neural network-based box office predictor and noted that different film types exhibit different predictive dynamics, but did not formally compare global versus persona-aware models. Choudhary and Kaur (11) proposed a combined clustering and classification pipeline for customer retention, though it lacked systematic ablation of whether segmentation added predictive value. Our work fills this gap with a three-experiment design: a global baseline (Experiment 1), a cluster-augmented global model (Experiment 2), and fully separate per-persona models (Experiment 3), evaluated on a consistent held-out test set across all configurations.

For interpretability, Lundberg and Lee (12) introduced SHAP (SHapley Additive Explanations) as a unified framework for model-agnostic feature attribution. SHAP has been widely adopted in churn analytics (13) and financial prediction (14) to explain complex tree-based models. We apply SHAP TreeExplainer both globally and within each persona cluster, producing feature importance comparisons that directly answer the insight component of our research question: whether different types of movies succeed for different reasons.

3 Methodology

This project combines clustering, supervised machine learning, and explainability techniques into a single pipeline for movie success prediction. The overall workflow consists of four stages: data preprocessing and feature selection, movie persona discovery through K-Means clustering, predictive modeling using three classifier families, and interpretability analysis using SHAP values.

3.1 Data Preprocessing

The raw dataset contained 1,191,754 entries from TMDB, filtered to retain only films with non-zero budget and revenue values and a minimum of 10 votes, yielding 11,496 usable samples. The binary success label was defined as revenue exceeding twice the production budget, yielding a near-balanced class distribution of 44.5% positive cases. Crew presence columns (`director`, `cast`, `writers`, `producers`) were encoded as binary presence indicators (0/1), capturing whether a film had a

recognized crew listed without introducing high-cardinality categorical variables. Genre fields were transformed using multi-label one-hot encoding, producing 18 binary genre columns. Numerical features including budget, popularity, and IMDb votes were log-transformed using `log1p` to correct for heavy right skew. All features were standardized using `StandardScaler` fitted exclusively on the training set to prevent data leakage. LassoCV with 5-fold cross-validation was applied to the training set for feature selection, reducing the feature space from 38 to 35 predictors by zeroing out the `has_cast` coefficient. The dataset was split 80/20 into training (9,196 samples) and test (2,299 samples) sets using stratified sampling to preserve class balance.

3.2 Movie Persona Clustering

K-Means clustering was applied to the scaled training features to group movies into interpretable personas. To determine the optimal number of clusters, both the Elbow Method and Silhouette Score were evaluated for $k = 2$ to 10.

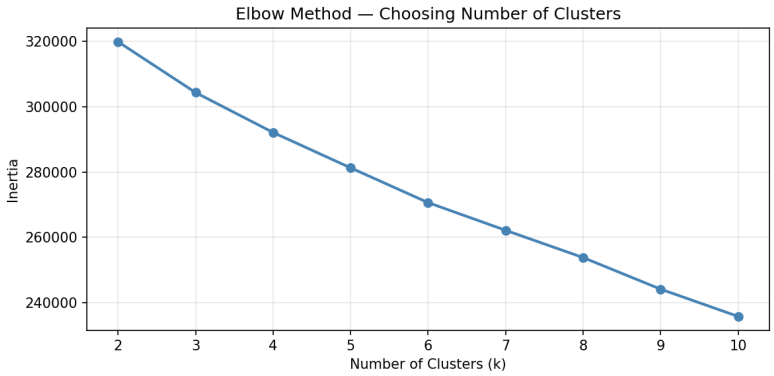

Figure 1: Elbow Method for selecting the optimal number of clusters. Inertia decreases sharply up to $k = 3$ before plateauing.

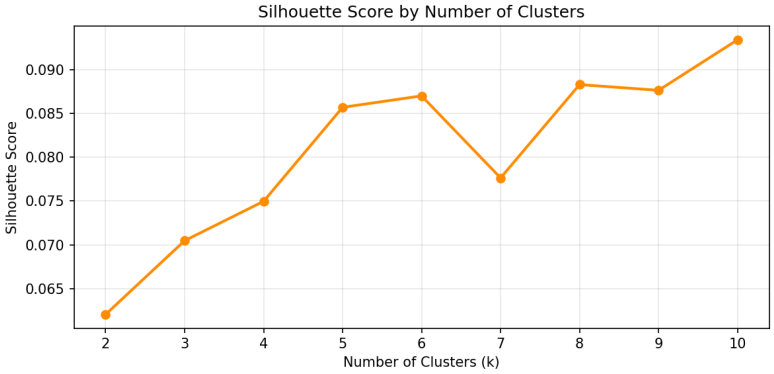

Figure 2: Silhouette scores by number of clusters. $k = 3$ was selected as the best balance between cluster compactness and interpretability.

Although silhouette scores increased slightly beyond $k = 3$, the gains were marginal (< 0.005 per additional cluster on average) and the elbow in the inertia curve identifies $k = 3$ as the point of steepest reduction before plateauing. Crucially, $k = 3$ maps directly onto recognizable industry categories, whereas higher values produced fragmented clusters without distinct interpretable profiles. Cluster assignments for the test set were generated using the fitted K-Means model to prevent data leakage. PCA was applied for two-dimensional visualization of the resulting clusters without influencing the clustering process itself. The three resulting personas were interpreted as:

- **Mainstream Blockbusters:** high budget, high popularity, large vote counts

- **Mid-Budget Commercial Films:** moderate engagement, balanced feature profile
- **Indie / Low-Visibility Films:** low budget, lower popularity, smaller audience reach

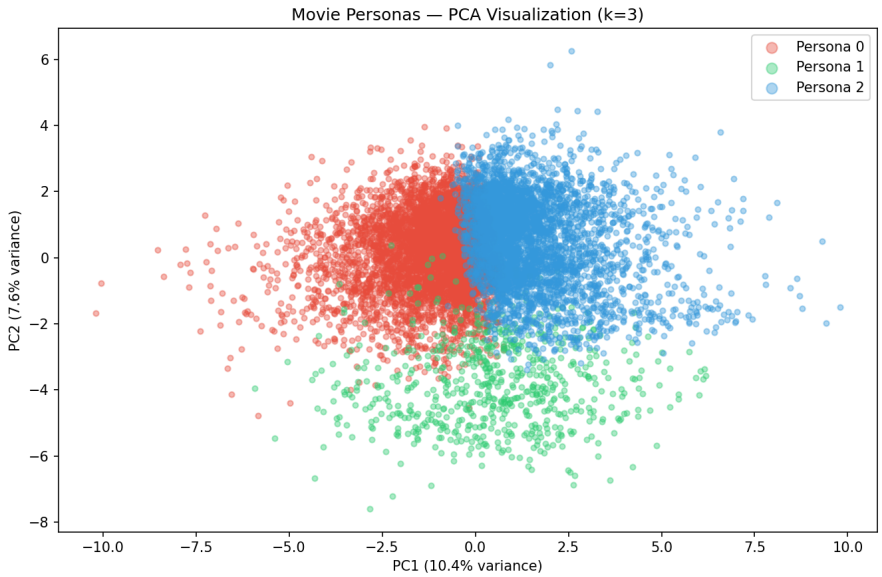

Figure 3: PCA visualization of K-Means clusters ($k = 3$). Each color represents one movie persona.

3.3 Predictive Modeling

Three supervised learning models were trained and evaluated across all experimental setups.

Logistic Regression was used as a linear baseline. It models the log-odds of success as a linear combination of input features:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}} \tag{1}$$

Random Forest is an ensemble of decision trees trained on bootstrap samples of the data. It reduces variance through averaging and provides built-in feature importance scores, making it robust to noise in tabular datasets.

Random Forest aggregates predictions from T decision trees, each trained on a bootstrap sample of the training data. The final prediction is determined by majority vote:

$$\hat{y} = \text{mode}\,\{h_1(x),\, h_2(x),\, \ldots,\, h_T(x)\} \tag{2}$$

where $h_t(x)$ is the prediction of the t-th tree. Feature importance is computed as the mean decrease in impurity (Gini impurity) across all trees and split points, providing a built-in ranking of predictive variables.

XGBoost trains trees sequentially in an additive fashion, where each new tree corrects the residual errors of the previous ones. The objective function is:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{m=1}^{M} \Omega(h_m) \tag{3}$$

where $l(y_i, \hat{y}_i)$ is the loss function and $\Omega(h_m)$ is a regularization term that penalizes model complexity to prevent overfitting (9). Three experimental setups were compared:

1. **Global Models:** Models trained on the full dataset with no clustering information.
2. **Global Models with Persona Labels:** Cluster assignments added as an additional feature.
3. **Persona-Specific Models:** Separate models trained independently for each persona.

3.4 Explainability with SHAP

SHAP (SHapley Additive Explanations) (12) was used to explain model predictions at both global and persona-specific levels. SHAP TreeExplainer was applied to the XGBoost models to compute feature attribution values, measuring how much each feature contributes to increasing or decreasing the predicted probability of success. Global SHAP analysis identified the most important features overall, while per-persona SHAP analysis compared how feature importance shifted across the three movie clusters.

4 Evaluation

4.1 Experimental Setup

The final processed dataset contained 11,496 movies after preprocessing and filtering. The dataset was divided into training and testing sets using an 80/20 split while maintaining class balance through stratified sampling.

All clustering experiments were performed using the training set only. Cluster assignments for the testing set were generated using the fitted K-Means model to prevent data leakage. Features were standardized prior to clustering and modeling to ensure that variables with larger scales did not dominate the learning process.

The main classification models evaluated in this project were Random Forest and XGBoost. Random Forest was selected because of its stability and ability to handle nonlinear relationships, while XGBoost was included because of its strong performance on structured datasets and ability to capture more complex feature interactions.

All models were trained using default hyperparameters (XGBoost: learning rate 0.3, max depth 6, 100 estimators; Random Forest: 100 estimators, unbounded depth; Logistic Regression: L2 regularization, $C = 1.0$), as the primary goal was comparing experimental configurations rather than maximizing absolute performance.

4.2 Evaluation Metrics

Model performance was evaluated using several standard binary classification metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Accuracy measured overall prediction correctness, while Precision and Recall were used to evaluate the balance between false positives and false negatives. F1-score provided a balanced measure between Precision and Recall, while ROC-AUC measured how effectively the models separated successful and unsuccessful movies across different classification thresholds.

4.3 Model Comparison

Table 1 summarizes performance across all seven model configurations. The global XGBoost model achieved the highest ROC-AUC of 0.837, outperforming all persona-aware configurations. The narrow performance gap of 0.017 across all configurations indicates that the engineered features carry sufficient predictive signal regardless of whether the model is segmentation-aware.

Table 1: Performance comparison across all experiments and models.

Model	Accuracy	F1	ROC-AUC
Exp1 – Logistic Regression	0.7434	0.6946	0.8205
Exp1 – Random Forest	0.7495	0.7070	0.8298
Exp1 – XGBoost	0.7521	0.7164	**0.8370**
Exp2 – Logistic Regression + Cluster	0.7434	0.6959	0.8207
Exp2 – Random Forest + Cluster	0.7534	0.7112	0.8319
Exp2 – XGBoost + Cluster	0.7486	0.7167	0.8355
Exp3 – Per-Persona XGBoost	0.7447	0.7104	0.8233

Table 2: Per-persona performance for Experiment 3.

Persona	Accuracy	F1	ROC-AUC
Mainstream Blockbusters	0.7515	0.6456	0.8131
Indie / Low-Visibility	0.7371	0.7195	0.8020
Mid-Budget Commercial	0.7376	0.7603	0.8232

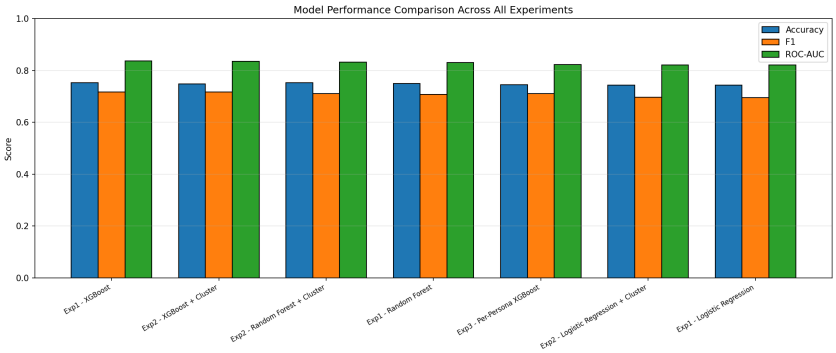

Figure 4: Performance comparison across all experiments and model families measured by Accuracy, F1, and ROC-AUC.

4.4 Interpretability Analysis

SHAP analysis revealed that audience engagement and budget-related features were the strongest global predictors of movie success. As shown in Figure 5, log IMDb votes (mean $|\text{SHAP}| = 1.07$) and log budget (0.94) were the two most influential features, followed by release year (0.46), vote count (0.40), and IMDb rating (0.37).

The per-persona SHAP analysis shown in Figure 6 revealed meaningful differences in feature importance across the three personas. Budget had a substantially stronger impact on Mainstream Blockbuster predictions (SHAP = 1.18) than on Indie/Low-Visibility films, where vote count dominated (SHAP = 1.45). Mid-Budget Commercial films showed the highest sensitivity to IMDb rating quality (SHAP = 0.46), suggesting that reception quality plays a disproportionate role for mid-tier productions. These differences confirm that movie success is not driven by the same factors across all film types.

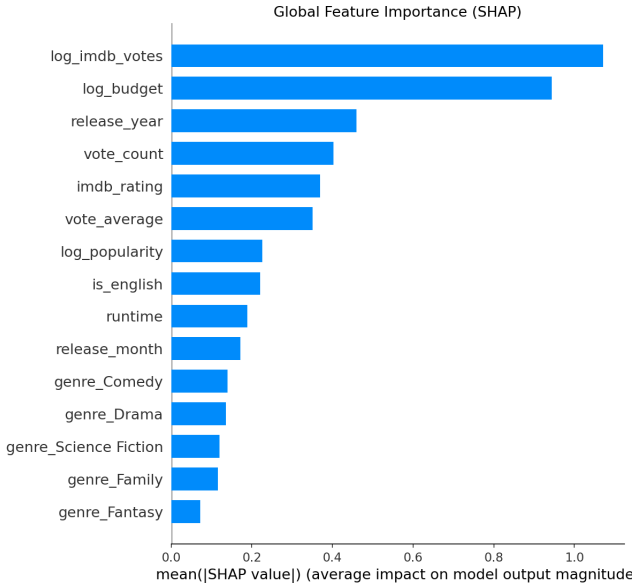

Figure 5: Global SHAP feature importance. Log IMDb votes and log budget are the two strongest global predictors of commercial success.

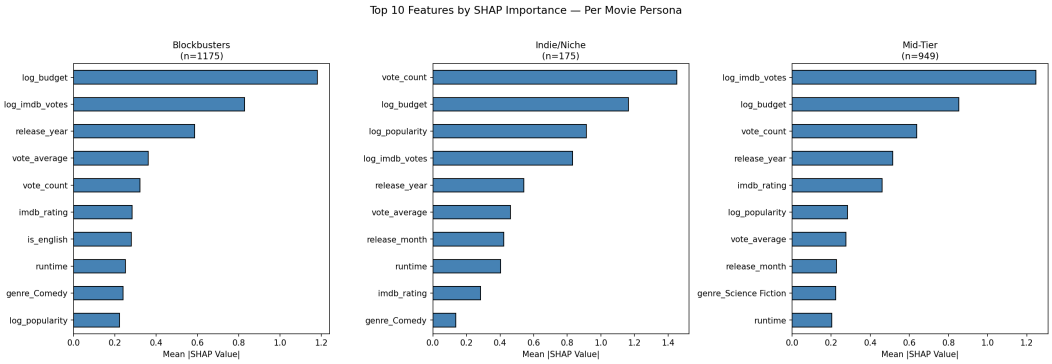

Figure 6: Per-persona SHAP feature importance across Mainstream Blockbusters, Indie/Low-Visibility, and Mid-Budget Commercial film types.

5 Discussion

5.1 Predictive Performance Across Experiments

Across all three experiments, XGBoost consistently outperformed Logistic Regression and Random Forest. The global XGBoost model (Experiment 1) achieved the highest overall ROC-AUC of 0.837, with accuracy of 0.752 and F1 of 0.716. Adding the cluster label as a feature (Experiment 2) produced only marginal changes in performance, suggesting that the cluster label largely captured information already present in the engineered features. The per-persona XGBoost models (Experiment 3) achieved a combined ROC-AUC of 0.823, remaining close to the global baseline.

The narrow performance gap across all configurations (0.017 ROC-AUC) suggests that the engineered features carry sufficient predictive signal even without segmentation-aware modeling. Overall, the global model remained sufficient for prediction accuracy, while clustering provided greater interpretability rather than substantial predictive improvement.

5.2 Feature Importance and Persona Heterogeneity

Although predictive performance remained relatively stable across experiments, SHAP analysis revealed meaningful differences in the drivers of success across movie personas.

Globally, the most influential features were log IMDb votes (1.07) and log budget (0.94), followed by release year, vote count, and IMDb rating. This suggests that audience engagement and production investment are the strongest overall indicators of commercial success.

Per-persona analysis revealed more distinct patterns. For Mainstream Blockbusters, budget was the dominant predictor, reinforcing the importance of large-scale production investment for commercial performance. For Indie and Low-Visibility films, vote count and popularity were the strongest predictors, suggesting that audience reach and word-of-mouth engagement play a larger role than spending power. Mid-Budget Commercial films showed higher sensitivity to IMDb rating quality, indicating that audience reception becomes more important for films operating between blockbuster and indie scales.

These findings support the primary insight of this project: even when a global model is sufficient for prediction, the factors driving movie success differ meaningfully across film types.

5.3 The Budget Effect and ROI-Based Success

The budget SHAP plots revealed a generally negative relationship between budget and predicted success across personas. This result is largely explained by the ROI-based success label used in this study. Since success required films to earn more than twice their production budget, higher-budget films faced substantially larger revenue thresholds, making commercial success inherently more difficult to achieve.

6 Limitations

Several limitations of this study should be considered when interpreting the results.

Missing financial variables. The TMDB dataset does not include marketing and promotional expenditure, which industry analysts consistently identify as a primary determinant of theatrical revenue for major releases. The relatively lower F1 score for the Blockbuster persona (0.646) likely reflects this gap. Future work incorporating marketing spend data from sources such as The Numbers or Nash Information Services could substantially improve persona-specific prediction for high-budget productions.

Success label definition. Our binary success label, defined as revenue exceeding twice the production budget, is a simplified proxy for commercial viability. It does not account for home video and streaming revenue, ancillary merchandise income, or theatrical release windows, all of which contribute meaningfully to a film's actual financial outcome. Different threshold choices would alter the class distribution and could affect the conclusions drawn about which features drive success.

Cluster stability and interpretability. K-Means clustering is sensitive to initialization and does not guarantee stable cluster assignments across runs. Although we fixed the random seed for reproducibility (`random_state=42`), the persona labels are descriptive interpretations of cluster centroid profiles rather than formally validated categories. Alternative clustering methods such as Gaussian Mixture Models or hierarchical clustering could produce different segmentations and warrant exploration in future work.

Temporal generalizability. The dataset spans several decades of film releases, and the economics of movie success have shifted substantially over time due to the rise of streaming platforms, changing audience behavior, and the emergence of franchise-driven production models. Models trained on this combined historical sample may not generalize well to films produced in the post-2020 streaming era. A temporally stratified evaluation would provide a more robust assessment of out-of-sample performance in the current industry environment.

7 Conclusion

This paper examined whether movie persona segmentation improves commercial success prediction and whether success drivers differ across film types. Using 11,496 TMDB films, we combined clustering, supervised machine learning, and SHAP interpretability analysis into a single pipeline. Results showed that the global XGBoost model achieved the strongest predictive performance (ROC-AUC 0.837), while persona-aware models performed similarly but offered greater interpretability. SHAP analysis revealed that budget is most important for Mainstream Blockbusters, audience reach drives Indie and Low-Visibility films, and rating quality plays a larger role for Mid-Budget Commercial productions. Overall, the results suggest that persona-based modeling provides more value for understanding movie success than for improving prediction accuracy alone.

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